

Performance Testing

Date	29 June 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Functional Testing
Type of Testing	Functional & Performance Testing
Target Module / API	Home Page, Prediction Module, Result Page
Test Environment	Windows 11, Visual Studio Code, Python 3.13, Flask
Test Date	29-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Test Input	Expected Outcome
1	Launch the Flask application	Open http://127.0.0.1:5000	Home page loads successfully
2	Open Prediction Page	Click Predict HDI	Prediction page opens successfully
3	Predict HDI using valid inputs	Country, Life Expectancy, Schooling, GNI, Internet Users	HDI score and category displayed correct

4	Test multiple predictions	Different valid input values	Stable and accurate predictions without errors
---	---------------------------	------------------------------	--

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Home Page Loading Time	< 2 seconds	0.2sec	pass	Loaded successfully
2	Prediction Response Time	< 2 seconds	0.8 sec	pass	Fast prediction
3	Model Accuracy	High	R ² Score = 0.9999	pass	Excellent prediction accuracy
4	Error Rate	0%	0%	pass	No errors during testing
5	Application Stability	Stable	Stable	Pass	Multiple predictions completed successfully

Step 4: Observations & Analysis

Key Findings

- The application launched successfully on the local Flask server.
- The prediction page accepted valid inputs correctly.
- The Linear Regression model accurately predicted the HDI score.
- The predicted HDI category (Low, Medium, High, Very High) was displayed correctly.
- The application remained stable while performing multiple predictions.

Bottlenecks Identified

- No significant bottlenecks were observed during testing.
- The application responded quickly for all tested inputs.

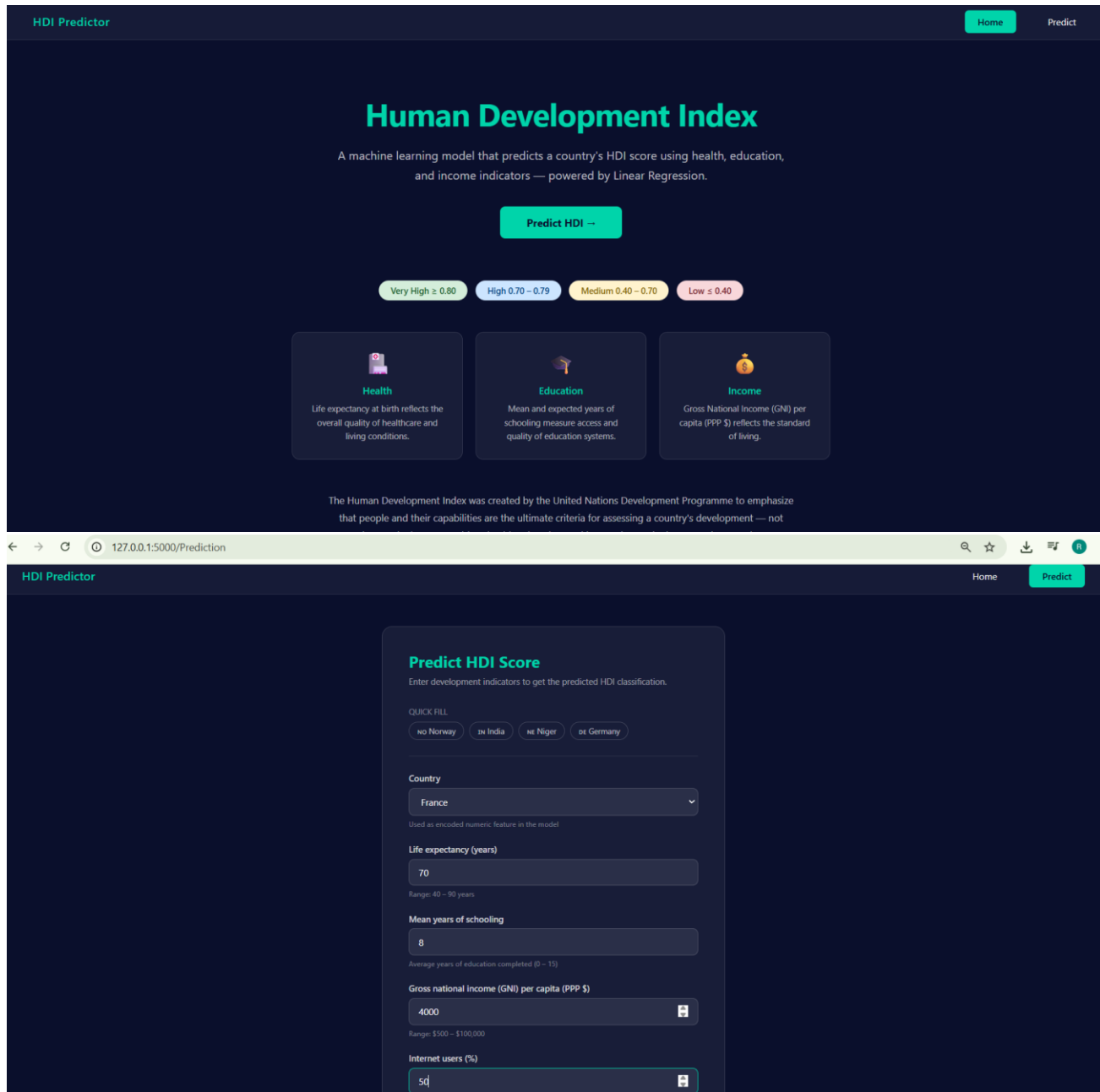
Optimization Steps Taken

- Trained the model using the Linear Regression algorithm.
- Saved the trained model using Pickle for faster loading.
- Loaded the trained model only once when the Flask application started.

- Tested the application with multiple valid input values to ensure reliable predictions..

Step 5: Screenshots / Evidence

The following screenshots present the performance testing results obtained using Apache JMeter.



The screenshot displays the HDI Predictor application interface. The top navigation bar includes "HDI Predictor" and "Home" buttons. The main heading is "Human Development Index", followed by a description: "A machine learning model that predicts a country's HDI score using health, education, and income indicators — powered by Linear Regression." A "Predict HDI" button is prominently displayed. Below this, four color-coded categories are shown: "Very High ≥ 0.80 " (green), "High $0.70 - 0.79$ " (blue), "Medium $0.40 - 0.70$ " (yellow), and "Low ≤ 0.40 " (red). Three informational cards are present: "Health" (Life expectancy at birth reflects the overall quality of healthcare and living conditions), "Education" (Mean and expected years of schooling measure access and quality of education systems), and "Income" (Gross National Income (GNI) per capita (PPP \$) reflects the standard of living). A footer note states: "The Human Development Index was created by the United Nations Development Programme to emphasize that people and their capabilities are the ultimate criteria for assessing a country's development — not".

The bottom screenshot shows the "Predict HDI Score" form. It includes a "QUICK FILL" section with buttons for "no Norway", "in India", "in Niger", and "in Germany". The "Country" dropdown is set to "France". The "Life expectancy (years)" input is "70" (Range: 40 - 90 years). The "Mean years of schooling" input is "8" (Average years of education completed (0 - 15)). The "Gross national income (GNI) per capita (PPP \$)" input is "4000" (Range: \$500 - \$100,000). The "Internet users (%)" input is "50".

HDI Predictor

Home

Predict


0.67
Medium Human Development

This country is in the medium development range.
Improvements in healthcare access, educational quality, and
income levels could significantly boost outcomes.

Predict again

Home